

Polish with Purpose: Surface Finish and Restoration Longevity



Group 4

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OUTLINE

BO1: Explain surface roughness, how it is measured, and its effect on the quality and glossiness of a composite restoration.

BO2: Describe the different steps for achieving ideal surface contour & smoothness. Identify the different types of smoothing systems used for posterior composites.

BO3: Describe the relationship between filler content and overall composite surface wear/longevity.

BO4: Describe the role of finishing and polishing systems in decreasing surface roughness of different composite types from immediate placement to ensuring long term smoothness and longevity.

BO5: Explain the maintenance strategies and repair techniques for composite restorations.

Case Studies + Decision Tree + Thinking Hats (Kahoot)

BO #1: Explain surface roughness, how it is measured, and its effect on the quality and glossiness of a composite restoration.

Josephine Widjaja

Search Strategy

PICO

P: Individuals with composite restorations

I: Surface smoothing (finishing/polishing)

C: No surface smoothing

O: Patient relevant consequences

Databases used: Pubmed, MDPI

Keywords: Composite resin, surface roughness

Mesh Terms: Composite resin AND surface roughness

Filters: 10 years, meta analysis, randomized controlled trials, systematic reviews

Search Results: 43 articles

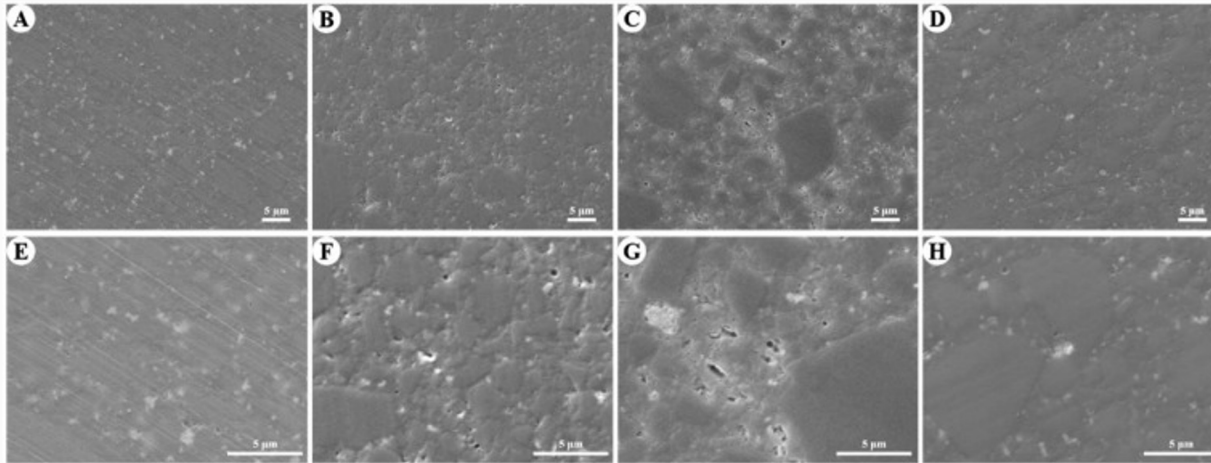
Highest Level of Evidence: Systematic Review

Query via ChatGPT 5:

"Please give me a the most commonly used measurements and their corresponding tools for surface roughness in peer-reviewed studies that look at the effects of roughness on patient relevant consequences of composite restorations"

Surface Roughness

Definition: **irregularities**, **deviations**, or **unevenness** on the material surface from its mean value due to cutting forces



Measurements of Roughness

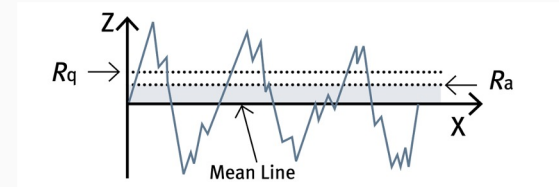
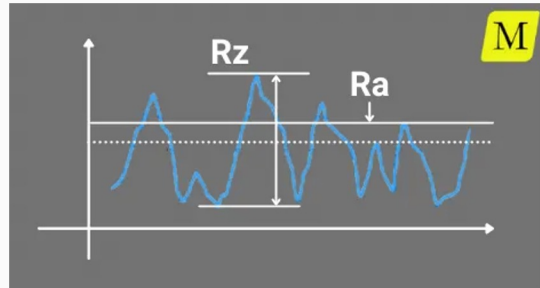
2D Profilometry (ISO 4287/4288) - *single line scan*

Ra value: measure of arithmetic average roughness (average absolute deviations of peaks/values from a mean line)

Rq value: root-mean-square average of profile heights

Rz value: average of highest five peaks and deepest five valleys

$$R_a = \frac{1}{L} \int_0^L |y(x)| dx$$
$$R_q = \sqrt{\frac{1}{L} \int_0^L [y(x)]^2 dx}$$
$$R_z = \frac{1}{5} \sum_{i=1}^5 (Z_{p,i} - Z_{v,i})$$



Measurements of Roughness

3D interferometry/confocal/AFM/CLSM (ISO 25178) - *entire surface area*

Sa value: 3D average deviation over area (~**Ra value over area**)

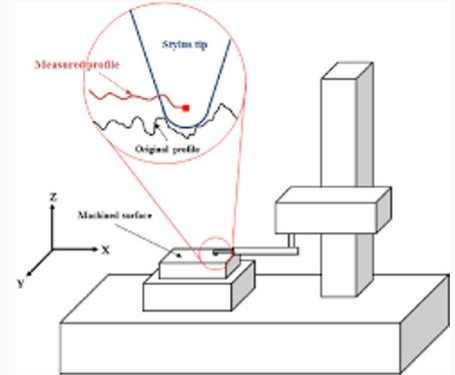
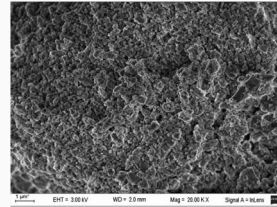
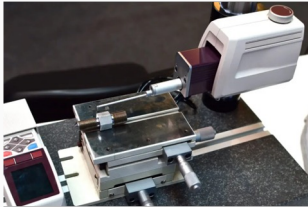
Sq value: root-mean-square average of profile height over area

Others

GU: gloss value, how well it reflects light (high gloss >70 GU, low gloss <10 GU)

Measuring Tools

- **Contact Stylus Profilometry** (gold standard for 2D values, Ra, Rq, Rz)
- **White-light interferometer/confocal microscope** (Preferred for 3D analysis, Sa, Sq +Ra)
- **Glossmeter** (GU)
- **Atomic Force Microscope** (Ra, Rq in nm)
- **Clinician Tactile Sense** (Ra threshold = **0.3 μ m**)
- **Scanning Electron Microscopy** (Qualitative)
- **Intraoral Scanners** (Not there yet!)

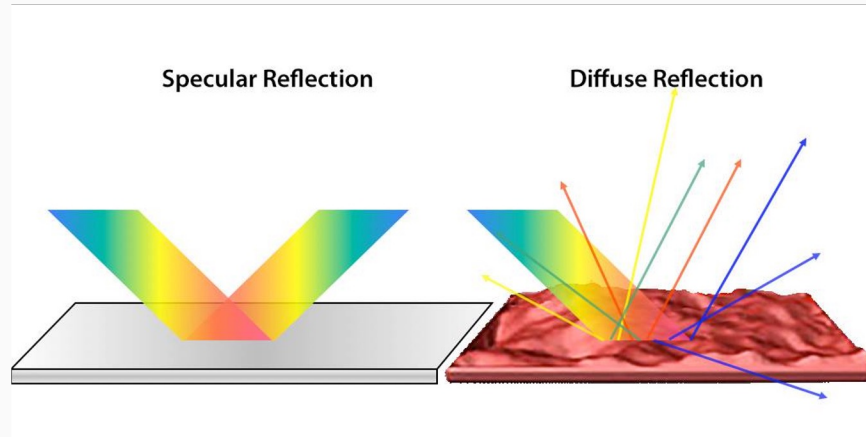


Effects of Surface Roughness

- Most articles point to roughness, only research related to smoothness correlate to dental implant surfaces demonstrating lower osseointegration (Łosiewicz B. et al., 2025)
- Roughness is shown to:
 - Increase dullness
 - Increase surface staining/color change/discoloration
 - Increase plaque/biofilm accumulation (**Threshold Ra = 0.2um**)
 - Increases abrasion/adhesion -> faster composite wear
 - Decreased microhardness
 - Decreased gloss
 - Decreased comfort for patient (**Threshold Ra = 0.25-0.5 um**)

More on Gloss

- **Specular Reflection** - appears glossy (**optically smooth limit: Ra 0.1 μ m**)
- **Diffuse Reflection** - appears matte/dull



BO #2: Describe the different steps for achieving ideal surface contour & smoothness. Identify the different types of smoothing systems used for posterior composites.

Jasmine Canlas and Brandon Ragone

Search Strategy

Databases used: **Pubmed**

MeSH Terms:

[finishing] OR [polishing] AND [dental composites]

[finishing] AND [burs] OR [discs] AND [dental composites]

[polishing] AND [burs] OR [discs] AND [dental composites]

Query via **ChatGPT**

"Please provide me with detailed information about the specific manufacturers, costs, and mechanism of action for each specific dental polishing technique that are used for posterior composites listed below. or each reference used, please make sure they are obtained from journals that are peer-reviewed and have high impact factors, and ensure that the article found is available in downloadable PDF format."

Highest Level of Evidence: Systematic Review

Finishing

finish \fin'ish\ *n* (1779): to put a final coat or surface on; the refinement of form prior to polishing

- Contouring
- Marginal refinement

vs.

Polishing

polishing \pöl'ish-ing\ *vb*: 1. to make smooth and glossy, usually by friction; to give luster to (GPT-1)

- Progressive abrasion
- Polishing pastes

Finishing & polishing achieves surface smoothness, which reduces plaque, improves wear resistance & marginal integrity, and enhances esthetics.



1. Finishing

(*macro-finishing*)

Purpose: Shape correction and contour refinement after restoration placement.

Key actions: Removal of gross excess, defining anatomy, margins, and proximal contacts.

Examples

Brasseler USA



Football Carbide
Flame Carbide

3M



Coarse
(Dark Orange)

Coarse Discs

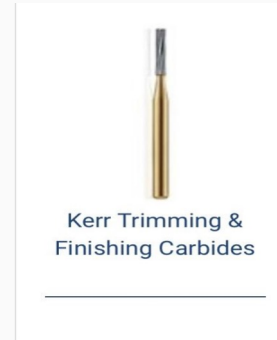
3M



Finishing Strips

Kerr® Finishing Burs

- Multi-fluted carbide burs available in fine and superfine grits
 - **8k-30k RPMs with water**
 - 0.3-1.0 μm Ra value
- A) Carbide burs have more blades that produce a finer cut with smaller scratches.
- B) For diamond burs, a higher number of diamond particles means each particle is smaller, also resulting in finer, less noticeable scratches.



**INTERPROXIMAL
BUCCAL
LINGUAL**



OCCLUSAL



In Vitro Study

Looked at how these sequences impacted *surface roughness* and hardness, comparing them to a Mylar strip control group.

Objectives:

- The aim of this lab study was to evaluate how two finishing-polishing sequences affect the surface of five nanotech-based resin composites.
- 1) The first sequence, QB, used a **12/15-fluted finishing bur** followed by an EVO-Light polisher.
- 2) The second, QWB, added a **30-fluted** polishing bur after the 12/15-fluted bur.

Results

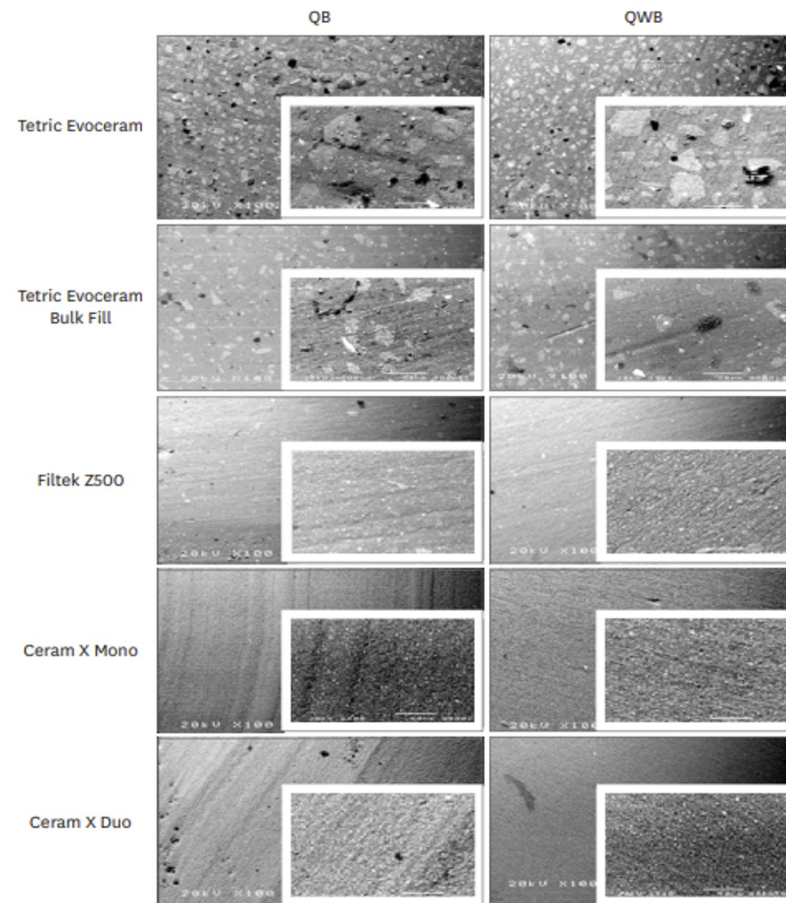
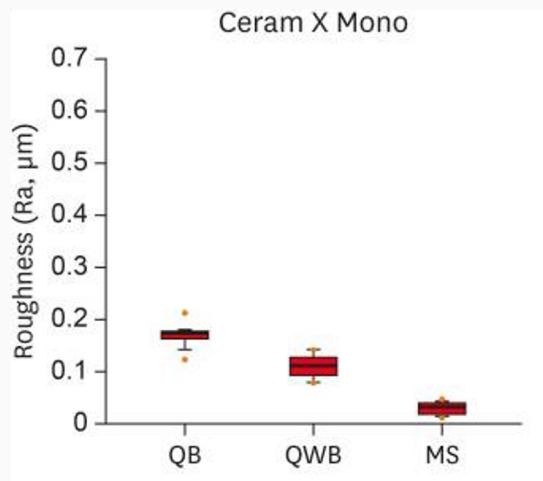
Conclusions:

- The **QWB sequence** (with the additional 30-fluted polishing bur) produced smoother surfaces and higher hardness than the QB sequence.
- Adding the intermediate polishing bur improved the surface finish across all five nanotech-based composites.

Significance:

- Smoother surfaces reduce plaque accumulation and staining, improving esthetics and longevity.
- Higher surface hardness means better wear resistance and durability of the restoration.
- Clinically, incorporating an additional fine-fluted bur step can significantly enhance the final quality of composite restorations.

SEM observations of 5 different nano-filled resin composites after 2 finishing-polishing sequences (QB and QWB)



Types of Abrasives

Abrasives for polishing

- 1) Silica – polishing pastes, smooth shine
- 2) Silicon carbide – shapes fast, leaves rougher surface
- 3) **Aluminum oxide** – discs/strips, smoother finish (Kerr Optidisc)
- 4) Diamond – hardest, final glossy polish

Why it matters

- Harder and finer = *smoother surface*
- Coarse → fine sequence *reduces scratches*
- Better esthetics and **longer-lasting** restorations

2. Polishing

(Lustration & gloss enhancement)

Purpose: Achieve a highly smooth, glossy, enamel-like surface.

Key actions: Final reduction of micro-roughness to $Ra < 0.2 \mu m$ (below bacterial adhesion threshold).

Examples

One-step

Dentsply Sirona



Enhance Points

Multistep

3M



Polishing Discs

Dentsply Sirona



Polishing Paste

Kerr® OptiDisc

- Flexible **aluminum oxide abrasives**
- Light pressure
- 0.1-0.2 um Ra value
- Color-coded discs
- With or without water



Kerr® ProGloss

- Rapid surface polish
- **Silicone**-based polisher with diamond abrasives
- Ideal for quick posterior touch-ups
- 0.1-.25 μm Ra value
- Wet or dry



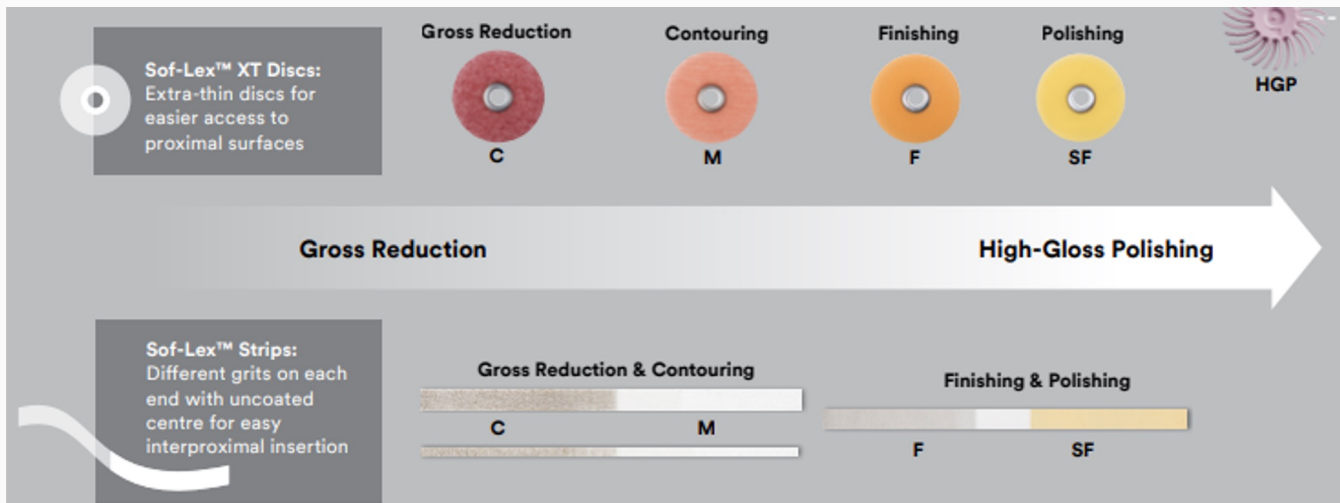
Disc: Facial and line angles
Points: Occlusal
Cups: Cusps, cervical line angles

Kerr® OptiShine

- Abrasive type: **silicon carbide** embedded in bristles
- Particle size: very fine, micron-sized
- Works by **direct contact** with the *composite surface*
- Mechanism: two-body wear (abrasive fixed to tool)
- Creates smooth, glossy finish **without separate polishing paste**
- Concave brush shape reaches hard-to-access areas



Finishing and polishing is a continuum of reducing abrasive sizes.



Grit/particle size

3M Sof-Lex Disc System

50-90 μm

Gross Reduction



C

10-40 μm

Contouring



M

3-9 μm

Finishing



F

1-7 μm

Polishing



SF

Brasseler USA Needle Carbide Bur



8-count blade



16-count blade

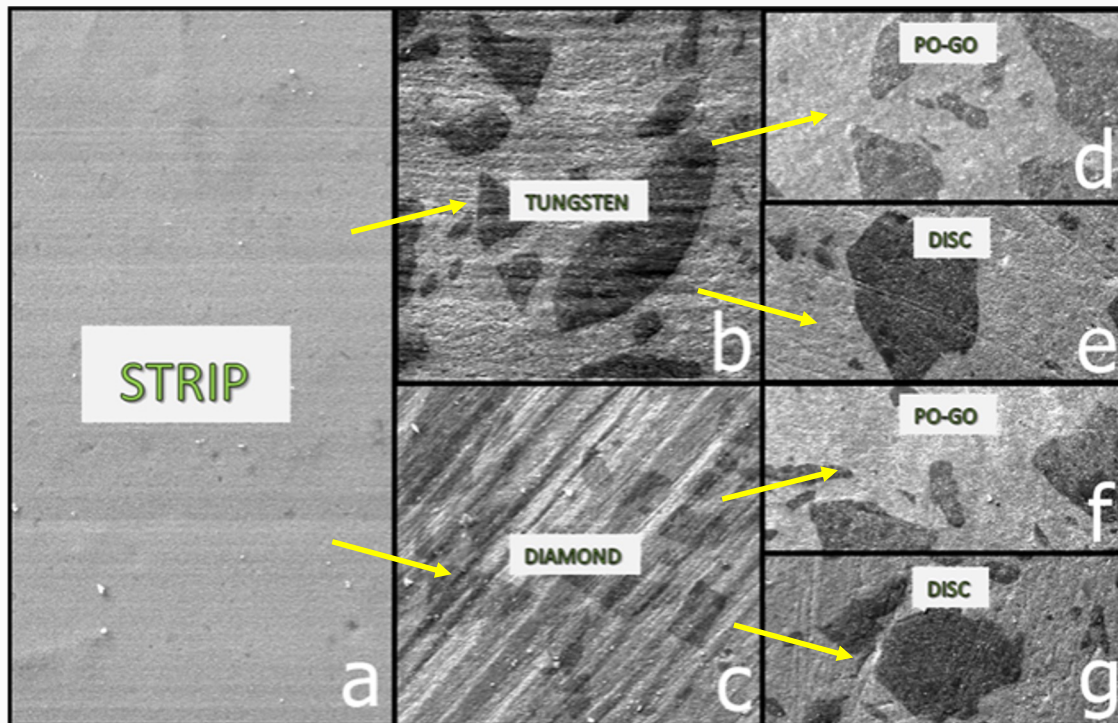


30-count blade

Blade Count

Comparison of One-Step and Multistep Polishing Systems for the Surface Roughness of Resin Composites

Ismail Hakkı Baltacıoğlu¹, Ozgur Irmak², Nuran Ulusoy³, Esra Cengiz³, Yıldırım Hakan Bağış¹



3. Surface Modification

(Glazing)*

Purpose: Optional post-polishing resin coating to seal microcracks and voids, improving gloss retention and wear resistance.

Examples

Light-cured



Low Viscosity
Liquid Polish
(Light-cured
glaze)



Nanofilled Self-
Adhesive Light-Cured
Protective Coating

Self-cured



Self-Cured Resin

*High-level research on dental glazes for composites is limited.

Bisco® Light Cured Glaze

- 1) Apply a thin layer of glaze over a finished and polished composite restoration
- 2) Low-Viscosity, flows into surfaces irregularities
- 3) Light Cure Polymerization
- 4) Improves gloss and smoothness

Self-cured Resin Glaze

- 1) Chemical set, not light cured
- 2) Not necessarily used for composite restorations
- 3) Mostly used on provisional appliances and reline materials

Does NOT replace polishing, used in combination!

Continued..

- In a 2009 in vitro study by **Cilli, et al**, they tested how surface sealants affect composite roughness under simulated toothbrushing
- Tested microhybrid and nanofilled composites
- Microhybrid: Unsealed and sealed surfaces had the same Ra values
- Nanofilled (Z350): Unsealed was smoother than the sealed ones

The materials used in this study are outlined in [Table 1](#).

Table 1. Materials used.

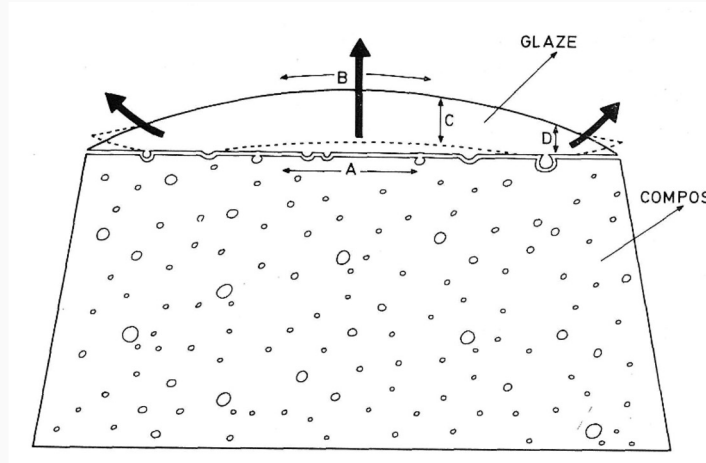
Materials	Manufacturer	Composition	Batch #
Sealant: BisCover	Bisco Inc.	Ethoxylated Bis-GMA, urethane acrylate, polyethyleneglycol diacrylate	0300012280
Sealant: Single Bond	3M-ESPE	Bis-GMA, HEMA, co-polymer of acrylic/itaconic acids, diurethane dimethacrylate, glyceroldimethacrylate, water and ethanol	7KK
Resin Composite: Z250	3M-ESPE	Resin: bis-GMA, UDMA, bis-EMA, TEGDMA; Filler (82wt%): zirconia/silica (size: 0.01–3µm)	6YP
Resin Composite: Z350	3M-ESPE	Resin: Bis-GMA, UDMA, Bis-EMA, TEGDMA; Filler (78.6wt%): non-agglomerated 20nm silica particles and agglomerated zirconia/silica nanoclusters (size: 0.6–1.4µm) consisting of agglomerates of primary zirconia/silica particles (size: 5–20nm)	7KB/7JK/7KF
Acid etchant: Etch-37	Bisco Inc.	37% phosphoric acid gel (polymer-thickened)	0500004535

BisCover is a one-step glaze applied directly to composites, while Single Bond plus Protect it! is a two-step adhesive and sealant system—but neither really improves measurable surface smoothness.

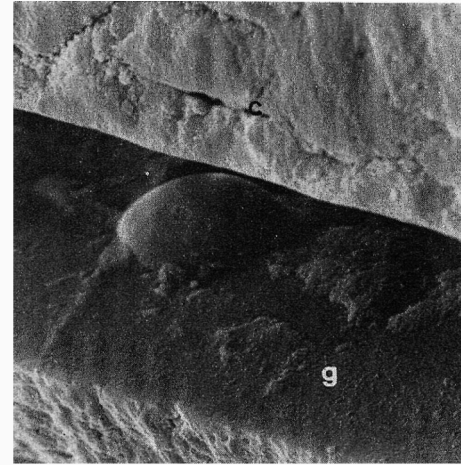
To glaze or not to glaze?

The glaze lies on the surface as a disc (Fig. 12). During polymerization the arc (B) undergoes a greater linear shrinkage than the chord (A). This results in **curling up of the margins**.

At the centre (C) the glaze layer is also thicker than at the margins (D) which results again in a **greater linear shrinkage** and a wider central interstice.



Lambrechts P., Vanherle G. (1982)



**BO #3: Describe the relationship
between filler content and overall
composite surface wear/longevity**

Kat Steinmetz

Search Strategy

P: Resin-based composite dental restorations

I: Composites with optimized filler characteristics (smaller, spherical, improved filler type (e.g., silica/zirconia blends), and higher filler loading)

C: Composites with less optimal filler characteristics (larger, irregular, lower filler loading, glass-based)

O: Enhanced surface wear resistance, polish retention, and restoration longevity

Databases used

Pubmed

Google Scholar

ChatGPT

MeSH Terms

("composite resin") OR ("dental composite") AND ("filler content") OR ("filler size") OR ("filler shape") AND ("surface roughness") OR ("longevity") OR ("wear resistance")

Filters: 10 years, meta analysis, randomized controlled trials, systematic reviews

Highest level of evidence

Systematic reviews

Before you finish/polish:



Kerr Harmonize



ceram.X



Ivoclar Tetric Series



Dentsply Sirona



3M ESPE Universal

Material in question:

Composite: Combination of multiple distinct parts or elements with different physical or chemical properties

- Goal: Achieve improved or tailored properties

Resin Composite:

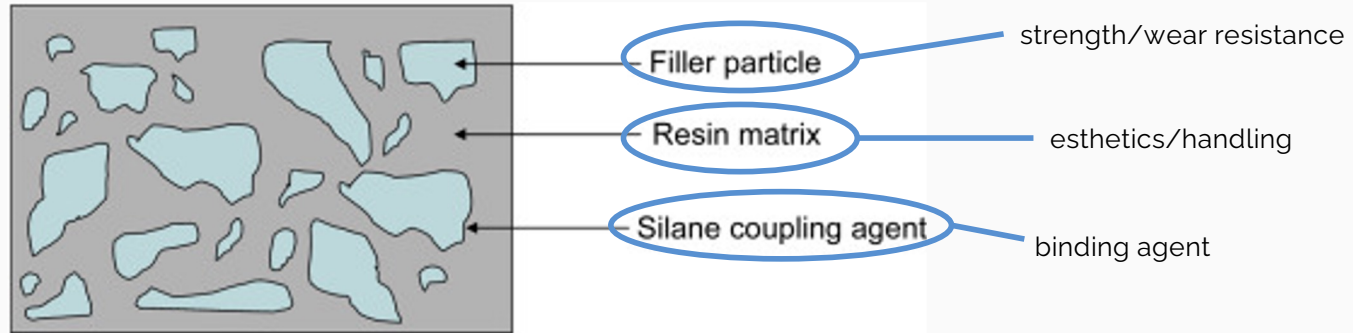


Image Source: Mirmohammadi, H (2018)

Key Filler Variables

Type:

- *Ground*: glass/quartz
- *Grown*: silica (SiO_2), zirconia (ZrO_2)

Shape:

- *Spherical*:
 - Better flow/handling
 - Even stress distribution
 - Slightly lower mechanical strength
- *Irregular/jagged*:
 - Greater mechanical strength
 - Stress concentration points
- Matters more for bulk mechanical strength and longevity than size alone (Rodriguez, 2019)

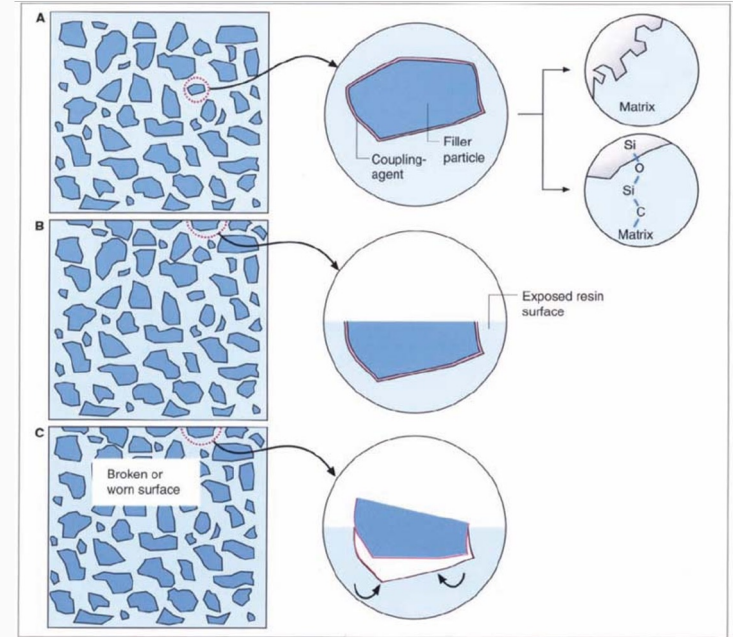


Image Source: PocketDentistry.com

Key Filler Variables Cont.

Size:

- Macro → micro → nano
 - Smaller filler → less surface irregularity over time, more esthetic
 - Larger filler → greater mechanical strength, less esthetic

Filler Loading (%):

- Volume or weight fraction of filler in resin matrix
- Most important factor for composite durability
- Higher loading:
 - ↑ strength, up to ~ 60 wt % (Rodriguez, 2019)
 - ↑ wear resistance
 - ↓ polishability (uneven abrasion between resin and filler)

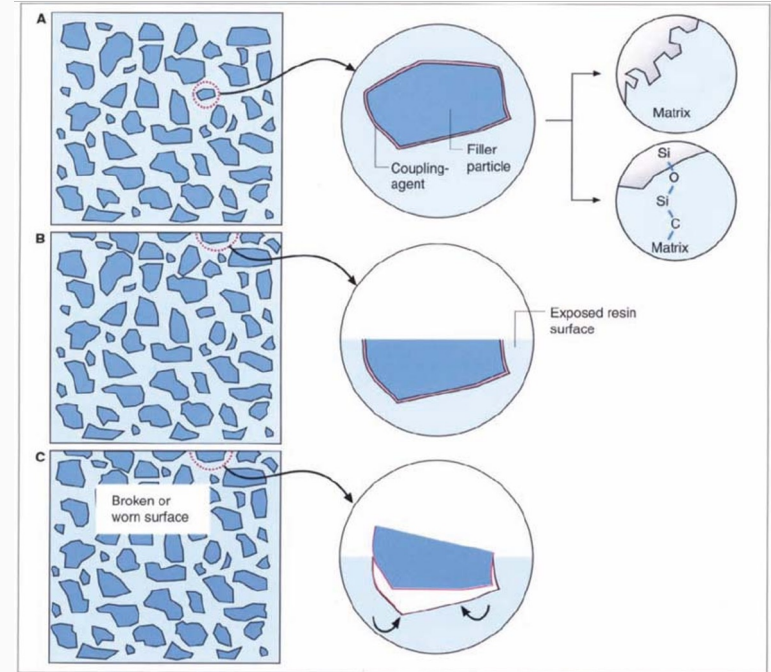
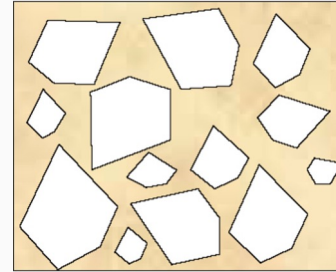


Image Source: PocketDentistry.com

Macrofilled Composites

Filler characteristics:

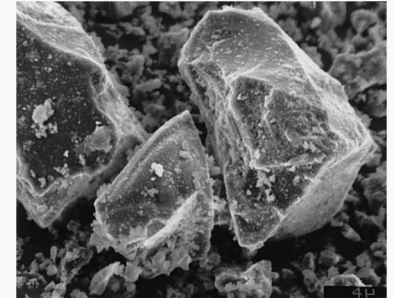
- **Type:** ground quartz or glass (radiopaque)
- **Shape:** irregular/jagged
- **Size:** 10 - 100 μm
- **Loading:** ~70 - 80 vol%



Ferracane 2011

Surface & Longevity:

- **Strength:** High compressive strength (filler loading)
- **Polishability:** *Poor* - large, irregular fillers protrude/dislodge during finishing
- **Gloss Retention:** *Extremely low* - filler “pluck-out” during wear → rough surfaces scatter light and accumulate plaque
- **Clinical Use:** Largely obsolete in modern practice

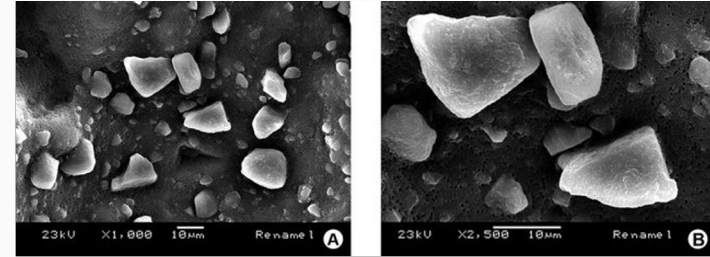


PocketDentistry.com

Microfilled Composites

Filler characteristics:

- **Type:** colloidal silica (SiO_2)
 - Chemically synthesized (“grown”) not ground
- **Shape:** spherical
- **Size:** 0.04 - 0.5 μm
- **Loading:** ~35 - 50 vol%



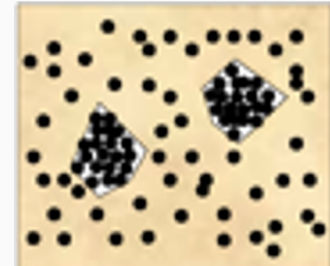
Berger 2009

Surface & Longevity:

- **Strength:** Lower than hybrids (low filler content)
- **Polishability:** *Excellent* - best initial gloss among all types
- **Gloss Retention:** *Excellent* - smooth, uniform filler distribution
- **Clinical Use:** Anterior restorations, veneers, cervical lesions



Renamel Microfill

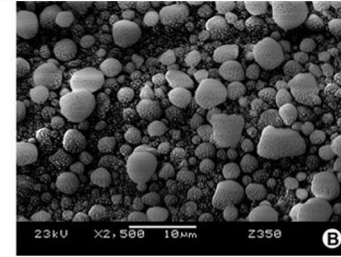
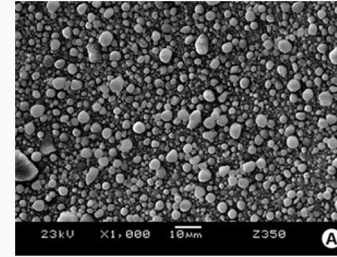


Ferracane 2011

Nanofilled Composites

Filler characteristics:

- **Type:** grown silica (SiO_2) and zirconia (ZrO_2) nanoparticles
- **Shape:** spherical (individual) and nanoclusters
- **Size:** 0.001–0.1 μm
 - High surface area-to-volume ratio $\rightarrow$ tend to agglomerate (solved by nanoclusters)
- **Loading:** ~60 - 70 vol%



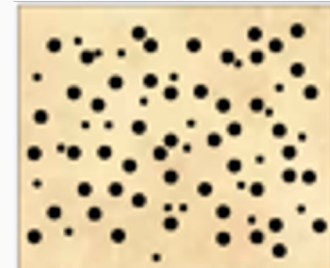
Berger 2009

Surface & Longevity:

- **Strength:** Comparable to hybrids (nanoclusters); more brittle
- **Polishability:** *Excellent* - smallest fillers
- **Gloss Retention:** *Excellent* - equal or slightly better than microfilled when clusters remain intact, minimal plucking
- **Clinical Use:** Anterior (gold standard) & posterior restorations



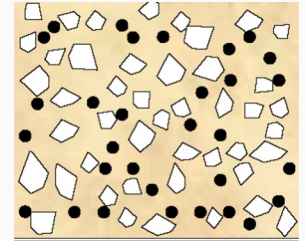
Filtek - Supreme Ultra
Universal Restorative



Ferracane 2011

Microhybrid Composites

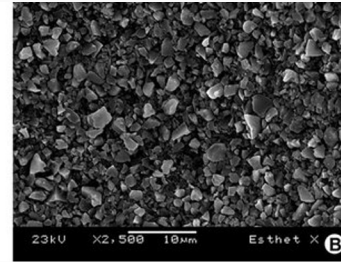
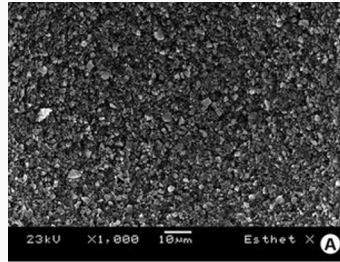
Hybrid = combination of smaller and larger particles



Ferracane 2011

Filler characteristics:

- **Type:** small silica particles + ground glass or quartz fillers
- **Shape:** mixed - spherical silica + irregular/jagged glass
- **Size:**
 - Silica: $0.04\ \mu\text{m}$
 - Glass/quartz: $0.4 - 1\ \mu\text{m}$
- **Loading:** ~55 - 65 vol%



Berger 2009

Surface & Longevity:

- **Strength:** *Very high*
- **Polishability:** *Good* - larger, more irregular particle size
- **Gloss retention:** *Moderate* - initial gloss achievable, but larger glass fillers can be lost during wear
- **Clinical Use:** Good for anterior and posterior restorations



Esthet-X HD Universal Composite

Nanohybrid Composites

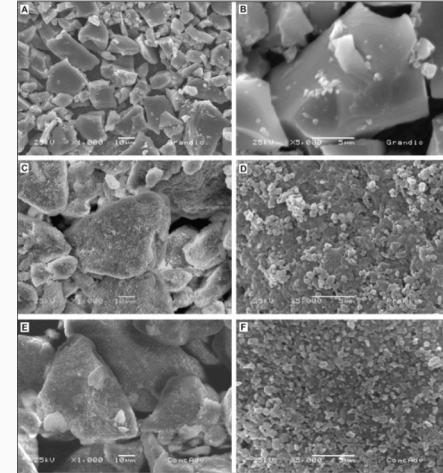
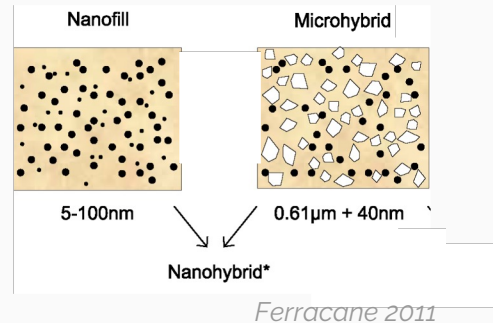
Hybrid = combination of smaller and larger particles

Filler characteristics:

- **Type:** silica/zirconia (grown nanoparticles) + barium glass (ground glass)
- **Shape:** mixed - spherical nanoparticles + irregular glass
- **Size:**
 - Nanoparticles: 0.02 - 1 μm
 - Submicron glass fillers: 0.4 - 0.6 μm
- **Loading:** ~60 - 75 vol%

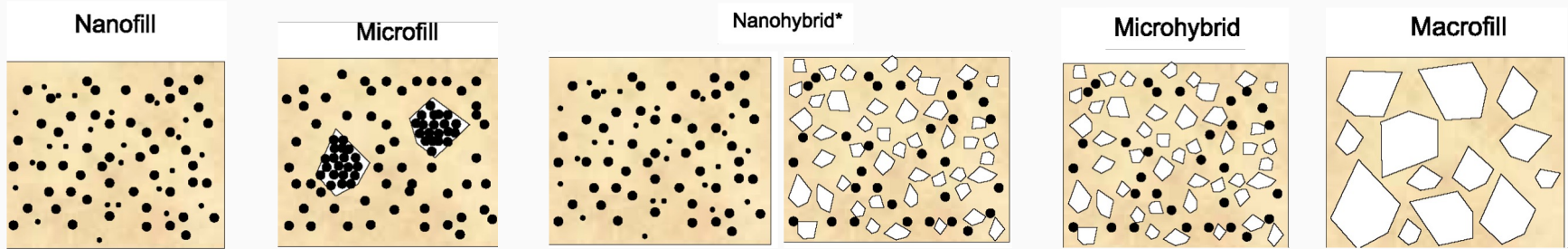
Surface & Longevity:

- **Strength:** *Excellent* (balanced mechanical/esthetic)
- **Polishability:** *Very good* - close to nanofilled
- **Gloss Retention:** *High* - improved resin-filler interface stability
- **Wear Resistance:** *Excellent* - reduced crack propagation and filler loss
- **Clinical Use:** True universal composite; ideal for both anterior esthetics and posterior strength



de Moraes 2009

Summary:



Filler size:

Nanofilled → Microfilled → Nanohybrid → Microhybrid → Macrofilled

Esthetics (polishability/gloss retention):

Macrofilled → Microhybrid → Nanohybrid → Microfilled ≈ Nanofilled

Durability:

Microfilled → Nanofilled → Nanohybrid → Microhybrid → Macrofilled

BO #4: Describe the role of finishing and polishing systems in decreasing surface roughness of different composite types from immediate placement to ensuring long term smoothness and longevity.

Hanan Dirirsa and Ashley Fordyce

Search Strategy

P: Resin-based composite restorations

I: Different finishing and polishing systems

C: Other finishing/polishing methods or unpolished controls

O: Surface roughness and gloss



Finishing and Polishing vs. Surface Roughness

Roughness Analysis on Composite Materials (Microfilled, Nanofilled and Silorane) After Different Finishing and Polishing Procedures

Open Access

Francesco Pettini¹, Massimo Corsalini^{1*}, Maria Grazia Savino¹, Gianluca Stefanachi¹, Daniela Di Venere¹, Carmine Pappalettere², Giuseppe Monno² and Antonio Boccaccio^{2,*}

¹Dental School - University of Bari, Piazza Giulio Cesare 11 - 70125, Bari, Italy; ²Department of Mechanics, Mathematics and Management, Politecnico di Bari, Viale Japigia 182, 70126 Bari, Italy



- Nanofilled: Ceram X (CX)
- Microfilled: Esthet-X (EX)
- Hybrid: Filtek Z250 (FZ)

A Comparative Analysis of Different Finishing and Polishing Devices on Nanofilled, Microfilled, and Hybrid Composite: A Scanning Electron Microscopy and Profilometric Study

¹Rishi D Yadav, ²Deepak Raisingani, ³Divya Jindal, ⁴Rachit Mathur



- Esthet-X HD → Microhybrid composite
- Ceram X Mono → Nanohybrid composite
- Filtek Silorane → Silorane-based microhybrid composite

Study 1

MATERIALS TESTED

Nanofilled: Ceram X (CX)



Microfilled: Esthet-X (EX)



Hybrid: Filtek Z250 (FZ)

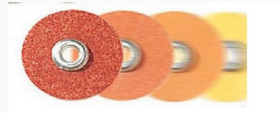


POLISHING SYSTEMS USED

Shofu Super Snap Rainbow Kit



Sof-Lex Pop-on Disks



Enhance System



METHODS

- 90 composite samples fabricated and cured
- Standard finishing with diamond bur (50 μ m)
- Surface roughness measured via Profilometer (Ra values) and SEM (5000 $\times$)

Results: Composite Type Comparison

Composite Type	Surface Smoothness (Polishability)	Key Reason
Ceram X (Nanofilled) 	Smoothest surface	Nanomer + nanocluster fillers reduce gaps → higher filler load & better polish retention
Esthet-X (Microfilled) 	Moderate smoothness	Contains BisEMA → softer resin matrix, slightly rougher
Filtek Z250 (Hybrid) 	Roughest surface	Large glass fillers plucked during polishing → voids & uneven texture

Results: Polishing System Comparison

Polishing System	Surface Smoothness (Polishability)	Key Reason
Shofu Super Snap 	Smoothest surface overall	Even abrasion of filler & resin; flexible aluminum oxide disks provide uniform polish
Sof-Lex Pop-on 	Smooth finish	Cuts resin & filler evenly; however, frictional heat may cause microcracks in resin
Enhance System 	Roughest finish	Abrades resin more than filler, leaving filler particles protruding and rough

Conclusions of Study 1

- **Nanofilled composite (Ceram X)** had the best polishability and smoothest surface.
- **Hybrid composite (Filtek Z250)** had the roughest surface after polishing.
- **Shofu polishing system** provided the smoothest finish across all materials.
- **Enhance system** resulted in the least smooth surfaces.
- Surface finish depends on both filler size/type and polishing system used.

Study 2

MATERIALS TESTED

Esthet-X HD
Microhybrid composite



Ceram X Mono
Nanohybrid composite

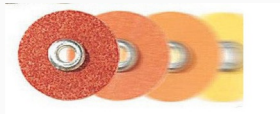


Filtek Silorane
Silorane-based microhybrid

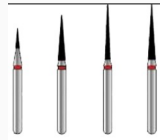


FINISHING & POLISHING TECHNIQUES USED

Sof-Lex abrasive discs



Diamond milling cutters



Tungsten carbide multi-blade cutters



Silicone rubber tips and polishing pastes

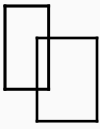


Results: Surface Roughness (Ra μm) of Different Composites by Finishing and Polishing Method

Group	Finishing / Polishing Method	Esthet-X HD (Microhybrid)	Ceram X Mono (Nanohybrid)	Filtek Silorane (Microhybrid)	Mean Ra (μm)	Observations
A	Control – no finishing or polishing	1.288	1.389	1.081	1.25	Roughest surfaces overall
BI	Finished with medium Sof-Lex discs	0.787	0.543	0.558	0.63	Moderate smoothness
B2I	Medium Sof-Lex + fine → superfine Sof-Lex polish	0.129	0.272	1.183	0.53	Very smooth for Esthet-X; rough for Filtek
B2II	Medium Sof-Lex + silicone rubber tip	0.355	0.528	0.126	0.34	Very smooth for Filtek
B2III	Medium Sof-Lex + Experience seal coat + polish paste (goat-hair + cotton mop)	0.328	0.529	1.055	0.64	Smooth for Esthet-X, rough for Filtek

Results: Surface Roughness (Ra μm) of Different Composites by Finishing and Polishing Method

Group	Finishing / Polishing Method	Esthet-X HD (Microhybrid)	Ceram X Mono (Nanohybrid)	Filtek Silorane (Microhybrid)	Mean Ra (μm)	Observations
C1	Diamond burs (50 μm → 30 μm) – no polish	1.556	1.444	0.469	1.16	Rough for hybrids, smoother for Filtek
C2I	Diamond burs + fine → superfine Sof-Lex polish	1.023	0.764	0.257	0.63	Moderate smoothness
C2II	Diamond burs + silicone rubber tip	1.318	0.712	0.535	0.86	Slightly rougher
C2III	Diamond burs + Experience seal coat + polish paste (goat-hair + cotton mop)	1.311	1.218	0.853	1.13	Rougher finish overall



Results: Surface Roughness (Ra μm) of Different Composites by Finishing and Polishing Method

Group	Finishing / Polishing Method	Esthet-X HD (Microhybrid)	Ceram X Mono (Nanohybrid)	Filtek Silorane (Microhybrid)	Mean Ra (μm)	Observations
D1	Tungsten carbide multi-blade cutters (Q + UF) – no polish	0.868	0.751	0.319	0.65	Smooth baseline
D2I	Carbide cutters + fine → superfine Sof-Lex polish	0.252	0.212	1.022	0.50	Smooth for Ceram X; rough for Filtek
D2II	Carbide cutters + silicone rubber tip	0.485	0.721	0.257	0.49	Very smooth for Filtek
D2III	Carbide cutters + Experience seal coat + polish paste (goat-hair + cotton mop)	0.475	0.455	0.342	0.42	Consistently smooth across all composites

Conclusions of Study 2

Smoothest combinations overall:

- **Esthet-X HD (microhybrid):** B2I (Sof-Lex fine/superfine)
- **Ceram X Mono (nanohybrid):** D2I (Carbide + Sof-Lex)
- **Filtek Silorane(silorane based microhybrid):** B2II (Sof-Lex + silicone rubber)

Roughest surfaces: Control (A) and diamond-only (C1) procedures ($R_a > 1 \mu\text{m}$).

Best overall finishing and polishing systems:

- Tungsten carbide multi-blade cutters (Q, UF)
 - Sof-Lex (fine / superfine)
 - Silicone rubber tip or Experience paste for final gloss
- Surface roughness is strongly influenced by the finishing/polishing sequence. Proper multi-step polishing can reduce R_a values by 70–90% compared with unpolished controls.**



Search Strategies

Chat GPT Query: "Please give me a PICO question and PubMed search strategy for studies examining how filler characteristics (such as filler content, size, morphology, or charge) affect the longevity and surface performance of resin-based composite restorations. The search should be broad enough to capture studies focusing on outcomes like surface roughness, gloss retention, wear resistance, surface degradation, or aging, and should include both in vitro and clinical investigations Format the answer with a PICO table and a PubMed-ready search string."

P: resin-based composite restorations

I: filler characteristics (content, size, morphology, charge)

C: different filler loadings/types

O: surface/long-term outcomes (roughness, gloss, wear, aging)

("Composite Resin" OR "Resin Composite" OR "Dental Composite Restoration")

AND

("filler content" OR "filler load" OR "filler particle size" OR "filler morphology" OR "filler charge" OR "filler particles")

AND

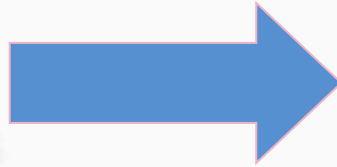
("surface roughness" OR "surface gloss" OR "gloss retention" OR "wear resistance" OR "surface degradation" OR "abrasion" OR "toothbrushing" OR "aging")

Longevity and Filler Content

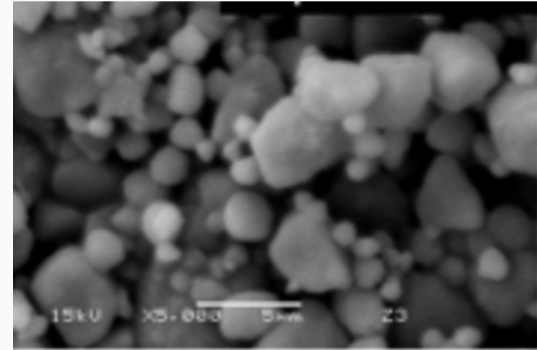
Polishing



Initial Surface
Smoothness



Filler Content



Longevity

Surface roughness and filler particles characterization of resin-based composites

Melissa A. Ruivo, Rafael R. Pacheco, Maicon Sebold ✉, Marcelo Giannini

First published: 16 July 2019 | <https://doi.org/10.1002/jemt.23342> | Citations: 45

Composite materials tested

2 conventional (Z100™, Filtek™ Supreme Ultra)

1 quick-cure (Estelite Σ Quick™)

1 fluoride-releasing (Beautifil II™)

1 self-adhering flowable (Vertise Flow™)

Dentifrice

Methods

Slide preparation Valo light cure for 20s; 5 composites for each group

Sof-Lex disc system for polishing with rinsing between each grit 15 - 20 s each

Simulated aging through storing in 37 degree C water for 1 week, brushed for 30,000 cycles (equivalent to 2 years of brushing), stored in water for additional 6 months. Organic content was removed to expose and evaluate filler particles

Surface roughness via profilometry and surface topography SEM filler composition evaluated by EDX microanalysis

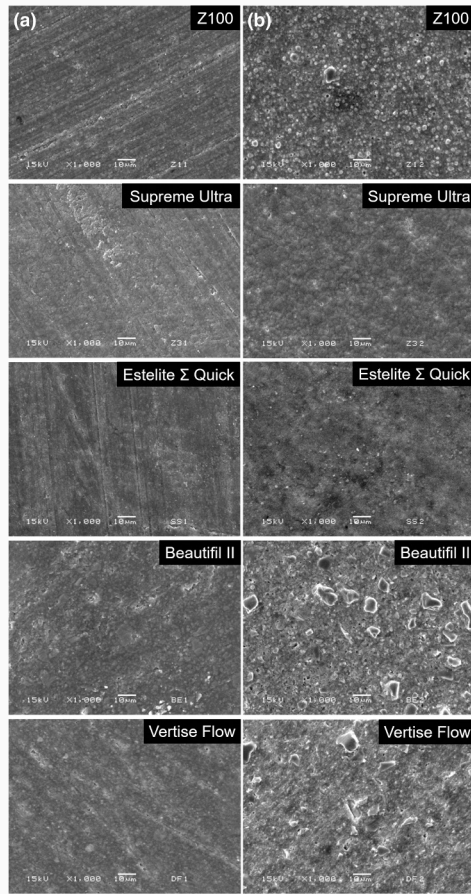


Composite Type	Filler Characteristics/vol%	Matrix/Features
Filtek Supreme Ultra (3M) (Nanofilled)	20 nm silica + 4–11 nm zirconia clusters (55.6–63.3%)	Silane treated ceramic, UDMA/ Bis-GMA
Z100 (3M) (Microhybrid)	0.01–3.5 µm silica-zirconia (66%)	Silane treated ceramic, Bis-GMA/ TEGDMA
Estelite Σ Quick (Tokuyama) (Supra Nanofilled)	200 nm spherical (71%)	RAP initiator
Beautifil II (Shofu) (Nanohybrid)	0.01–4 µm irregular S-PRG glass (68.6%)	Fluoride release
Vertise Flow (Kerr) (Microhybrid)	10–40 nm–20 µm irregular Al–Ba–Yb (48%)	GPDM, HEMA

Material	Ra (μm) Initial	Ra (μm) Aged	Roughness $\uparrow$ (*)	Longevity Interpretation
Filtek Supreme Ultra	0.29	0.74	2.6×	Smoothest → best long-term polish
Z100	0.35	1.09	3.1×	Good wear resistance
Estelite Σ Quick	0.25	1.12	4.5×	Acceptable short-term durability
Vertise Flow	0.55	2.88	5.2×	Degrades moderately fast
Beautifil II	0.49	3.44	7.0×	Highest roughness → poorest longevity

One week no brushing = no statistical differences in surface roughness

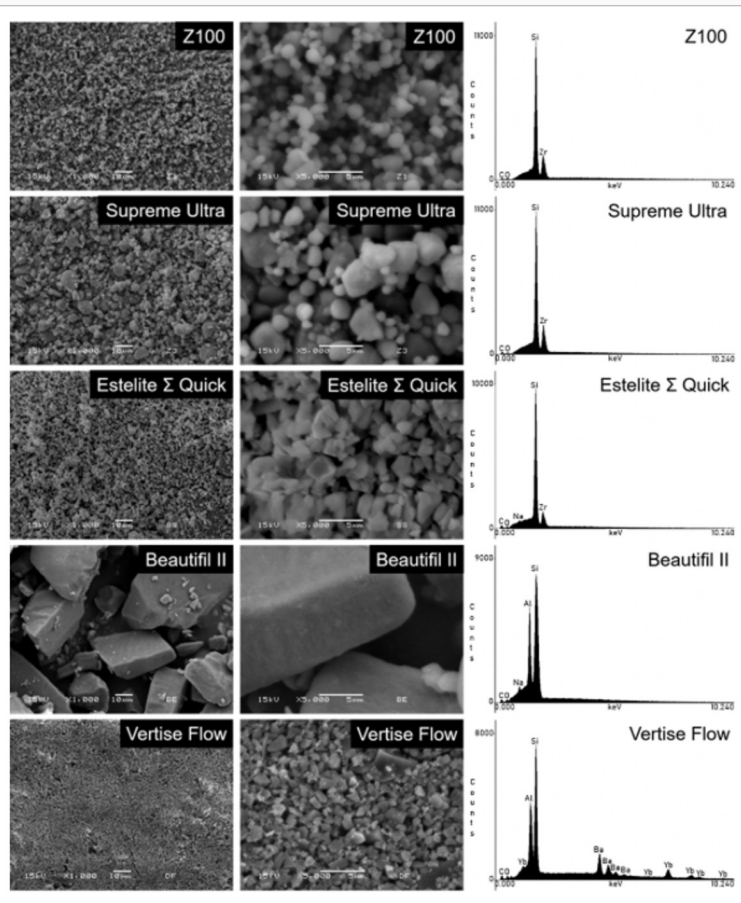
After brushing and 6 months of water submersion all increased in surface roughness with **Nanofilled & microhybrid composites** showing best polish retention



Surface topography after tooth brushing and water storage for 6 months

Abrasion removed resin matrix, exposing the filler leading to crater formation and the water sorption of the material leads to further degradation

Hydrophilic monomers (Bis-GMA, TEGDMA, HEMA and GPDM increase water absorption while hydrophobic (UDMA, BisEMA) monomers are more resistant



Irregular fillers had greater frictional forces and wore faster than spherical

Small spherical and rhomboid fillers 1-5 μm of silica and zirconia showed enhanced wear resistance

Findings

High Longevity / Low Maintenance

Filtek Supreme Ultra (Nanofilled): ideal for anterior + posterior esthetics

Z100 (Microhybrid): reliable for posterior high-stress areas

Moderate Longevity

Estelite Σ Quick: rapid polymerization

Low Longevity / High Maintenance

Beautifil II (Giomers): fluoride release beneficial but roughens quickly

Vertise Flow (Self-Adhering): roughness would be preceivable by patients due to
Ra = 0.55, hydrophilic; best used as liner/base

Summary

- Polishing ensures initial smoothness, but filler content and morphology dictate long-term surface durability.
- Smaller, spherical fillers + high filler load → lower roughness increase, better gloss retention → longer service life.
- Large or irregular fillers + hydrophilic matrices → increased roughness/gloss loss → more frequent maintenance.
- Clinical implication: Material selection should prioritize filler characteristics in addition to finishing/polishing technique.

BO #5: Explain the maintenance strategies and repair techniques for composite restorations.

Bren Kain and Seeta Patel

Search Strategies

Maintenance:

Databases: PubMed, Wiley, and The Journal of Professional Excellence: Dimensions of Dental Hygiene

MeSH Terms: (Composite) and (Maintenance) and (Longevity)

Highest Level of Evidence: Systematic Review

Repair:

Query via ChatGPT

"Please provide me with detailed information about repair techniques of abraded/ cracked composite restorations. For each reference used, please make sure they are obtained from journals that are peer-reviewed and have high impact factors, and ensure that the article found is available in downloadable PDF format."

Search Results: 5 articles

Highest Level of Evidence: Systematic Review

What are we maintaining?

Esthetics
Function
Integrity



Rodolpho, et al., 2022

Who is responsible?



Dentist



Dental Hygienist



Patient

Dental Hygienist's Role

- **Biannual Prophylaxis Appointments**

- Evaluating current restoration state
- Finishing & Polishing current restorations

- **Instrumentation Precautions**

- Ultrasonic Instruments
- Air-powder Polishing
- Rubber Cup Polishing

- **Professional Fluoride Application**

Alteration of composite resin after the application of fluoride varnish: an in vitro study

October 2019 · Journal of Dental Health Oral Disorders and Therapy · 10(5):305-308

DOI: [10.15406/jdhodt.2019.10.00503](https://doi.org/10.15406/jdhodt.2019.10.00503)

Varnishes Tested

- 1. Sodium Fluoride
- 2. NaF + Tri-Calcium Phosphate
- 3. NaF + Amorphous Calcium Phosphate + Casein Phosphopeptide

Methods

144 specimens of composite resin were prepared and polished.

Stored in artificial saliva, then varnish applied for 1 minute, every week for 4 weeks.

Results & Discussion

The varnish with sodium fluoride and tri-calcium phosphate generates greater roughness than sodium fluoride and sodium fluoride with amorphous calcium phosphate plus casein phosphopeptide.

Week	NaF	NaF + Tri-Calcium Phosphate	NaF + ACP+ CPP
1	0.30	0.30	0.30
2	0.31	0.33	0.31
3	0.32	0.35	0.32
4	0.33	0.42	0.33

estimated surface roughness values made using ChatGPT

Patient's Role

Oral Hygiene

- Toothbrush & toothpaste
- Technique
- Mouthwash
 - In Vitro Study

Caries Risk

- Diet
- Socioeconomic factors
- Access to care



Summary of Maintenance

- Dentists, dental hygienists, and patients are responsible for maintaining the esthetics, function, and integrity of their composite restorations.
- Dentists are responsible for informing their staff of the materials and importance in following the manufacturer's instructions.
- Dental hygienists are responsible for periodically evaluating the current state of the restorations and using the appropriate instruments and techniques to polish composite restorations for best possible long-term success.
- The patient is responsible for maintaining excellent oral hygiene using the appropriate products for the state of their oral environment. They are expected to understand their level of caries risk, the factors that influence that level of risk, and ways to lower their current risk.

To Repair or to Replace?

- When deciding whether to repair or leave an existing composite restoration, here are some factors to consider:
 - Time
 - Resources
 - Amount of tooth structure
 - Longevity of the restoration
 - Pulpal considerations
- Repair > replacement when:
 - Preserves tooth structure
 - Allows for longevity of the restoration
 - Reduced pulpal effects
- posterior teeth with composite restorations→ best monitored without intervention until there is evidence of plaque accumulation (Blum et. al)

CONTRAINDICATIONS FOR REPAIR

- Patient reluctance
- High CRA
- Secondary caries

Current Repair Techniques

AIR ABRASION

- Composite repair based on sandblasting with silica coated particles
 - 1. Sandblast resin substrate→ increased SA for bonding
 - 2. Apply silane agent
 - 3. Apply adhesive
 - 4. Place composite in 2 mm increments to repair defect

CONVENTIONAL ADHESIVE BONDING SYSTEM

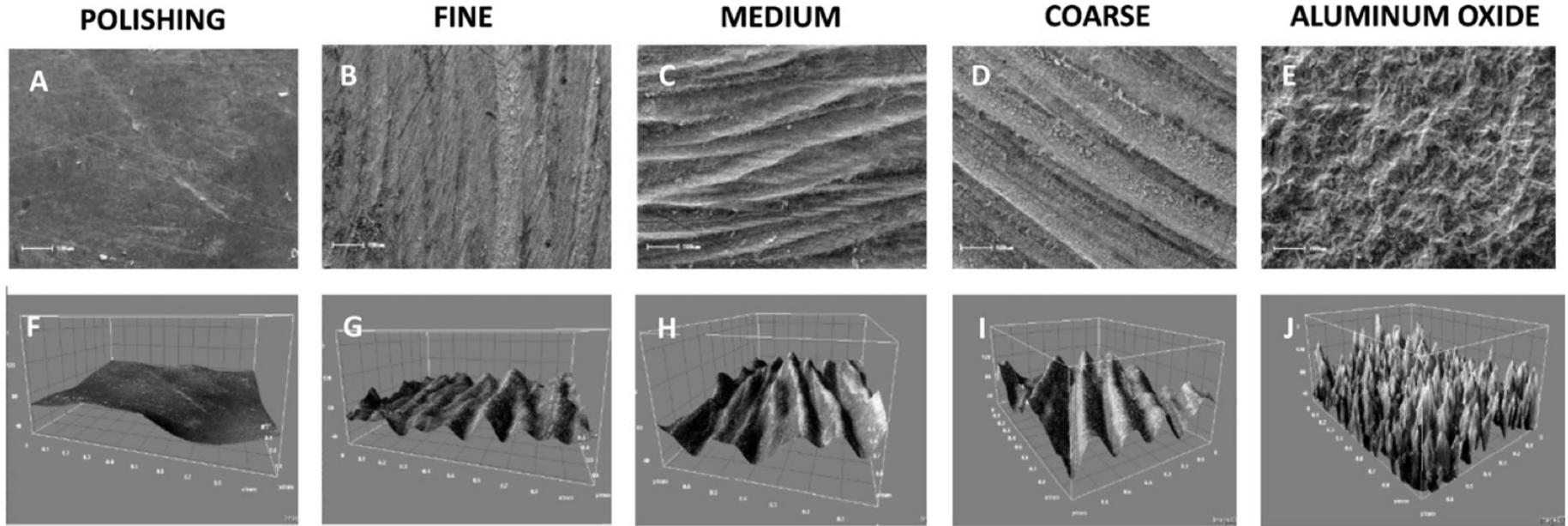
- 1. Acid etch 15-30 seconds
- 2. Apply adhesive system
- 3. Place 2 mm increments of composite and light cure



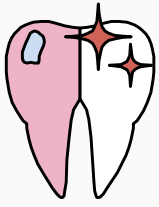
The BEST technique?

- *Durability of composite repair using different surface treatments (Feirerra da Costa et. al, 2012)*
 - 5 methods to improve bonding were studied:
 - Air abrasion with Aluminum Oxide (AO)
 - Chemical treatment with phosphoric acid
 - Chemical treatment with HF
 - Silane application
 - Surface roughening with diamond burs
- Results
 - Highest composite repair (CR) strength for AO group
 - SEM showed increased 3D roughness with AO compared to unidirectional roughness for diamond grit bur group
 - Lowest CR with untreated composite block group
- **Micro Retentive interlocking is the most important factor in establishing a good bond between old and new composite!**

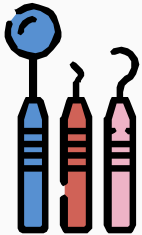
SEM images of surfaces after treatment



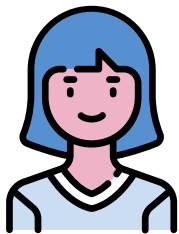
- *Feirerra da Costa et. al, 2012)*



Finishing & Polishing Our Decision-Making



Case #1



Katy Perio

Age	40
Gender	Female
Med Hx	Non-contributory
Dental TX	Direct composite resin veneer restoration on #8&9
CRA	Low

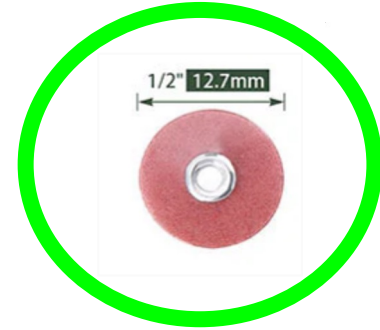


FIG. 4

Vargas et. al, 2022

Here is what your direct composite veneers look like immediately after applying composite veneers

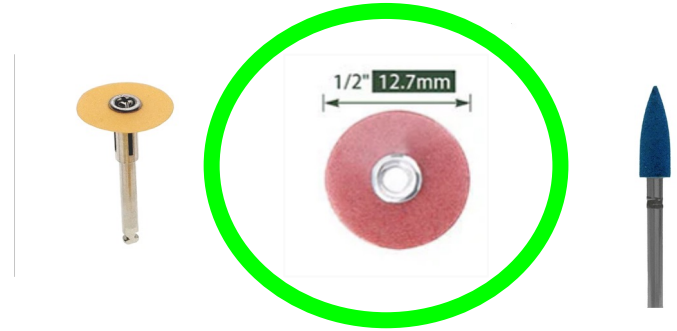
How should we adjust
the incisal edge?



****REMINDER****

Coarse discs are best
used for gross
contouring/adjustment.

How should we adjust
the incisal edge?



****REMINDER****

Coarse discs are best
used for gross
contouring/adjustment



FIG. 5

Which option would be the best to adjust facial contours and line angles?



****REMINDER****

Flame shaped burs have a tapered shape that makes it ideal for adjusting contours.

The pinpoint tip is very good for getting into tighters spots on the teeth.



Which option would be the best to adjust facial contours and line angles?

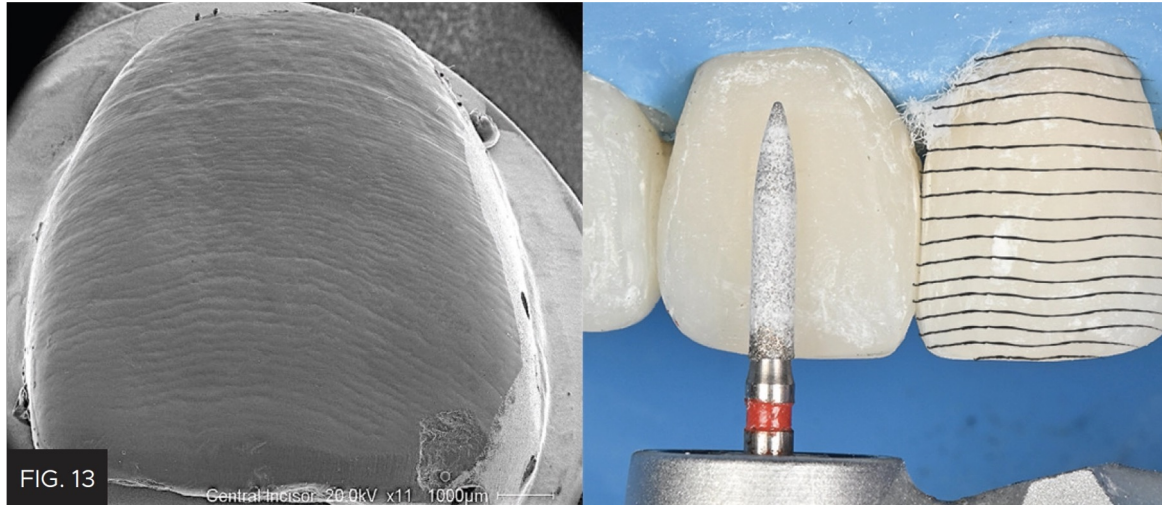


Flame
finishing
diamond bur



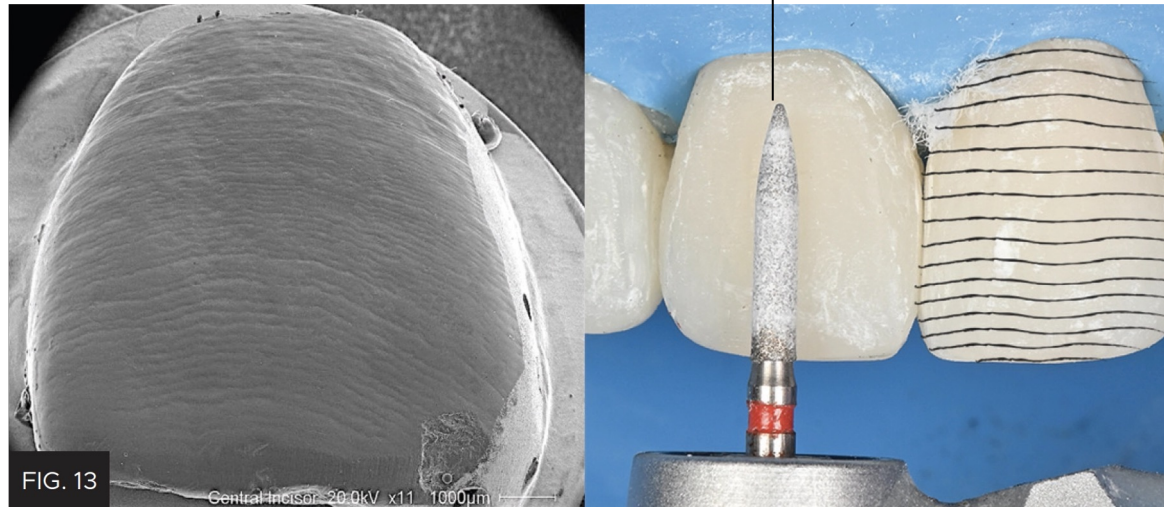
Now that the contours
have been established,
what step is next?

Fine-finishing



How does the SEM image and lines drawn on the tooth help us complete fine-finishing?

1. **Helps smooth surface irregularities**
2. **Creation of anatomical texture like perikymata and imbrication lines**



Now, onto the last step... **Polishing**



Why was the gray wheel selected?

- The case study only photographed one wheel being used; however, they claimed to use many different wheels in **sequential decreasing order of grits**.
- Each following wheel has **finer grit** and is **more flexible/soft**, resulting in the strongest shine! WHY?
 - Slow sequential removal of scratches and the softer wheels can have a burnishing effect



What could we use to help achieve the final shine?



****REMINDER****

Good: **One-Step Gloss Systems**

Better: **Multi-Step Gloss System**
since progressive refinement can help reduce the scratches on the composite.

Best: **Diamond Paste** (after the multi-step gloss system). The diamond particles burnish the composite to create more "Real-life" or enamel-like gloss.

What could we use to help achieve the final shine?



In this case, a 2-step
diamond polishing
system was used





FIG. 4

**Directly after curing
composite**



After finishing & polishing



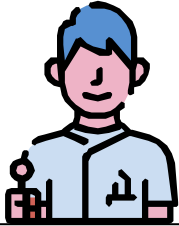
FIG. 15



FIG. 16

Case #2

Oral Health Group, 2021



Plaque Efron

Age	65
Gender	Male
Med Hx	Non-contributory
Dental TX	#12 DO (Class II Restoration)
CRA	High



Here is what your Class 2 restoration looks like directly after curing the composite.

First, check the bite
with articulating paper

First, check the bite
with articulating paper



Once the occlusion has been adjusted, we can begin working on creating proper anatomy.

Needle shaped
composite finishing
diamond bur



What is a good option to complete the polishing step?

****REMINDER****

It is important to use wheels in sequential order of decreasing grit and firmness to achieve the best shine.

Spiral Wheel used with an abrasive pre-polisher

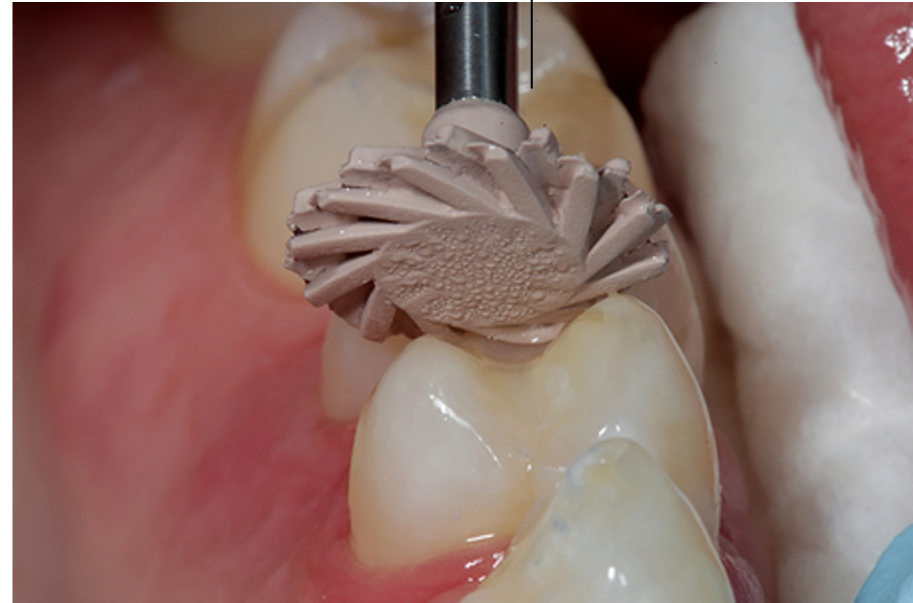


What could we do to complete the polishing step?

****REMINDER****

Wheel colors represent different levels of grit or abrasion. Between brands, there is no universal standard of colors in comparison to grit level. Be sure to read the label from the company you use.

Spiral Wheel used with an abrasive high shine polisher



What is the purpose of this instrument used?

A **composite saw** is used when the interproximal contact is too heavy (i.e. the floss breaks when forced through the contact).



What is the purpose of this instrument used?

A **composite saw** is used when the interproximal contact is too heavy (i.e. the floss breaks when forced through the contact).

To further smooth the interproximal area, consider using **finishing strips**.





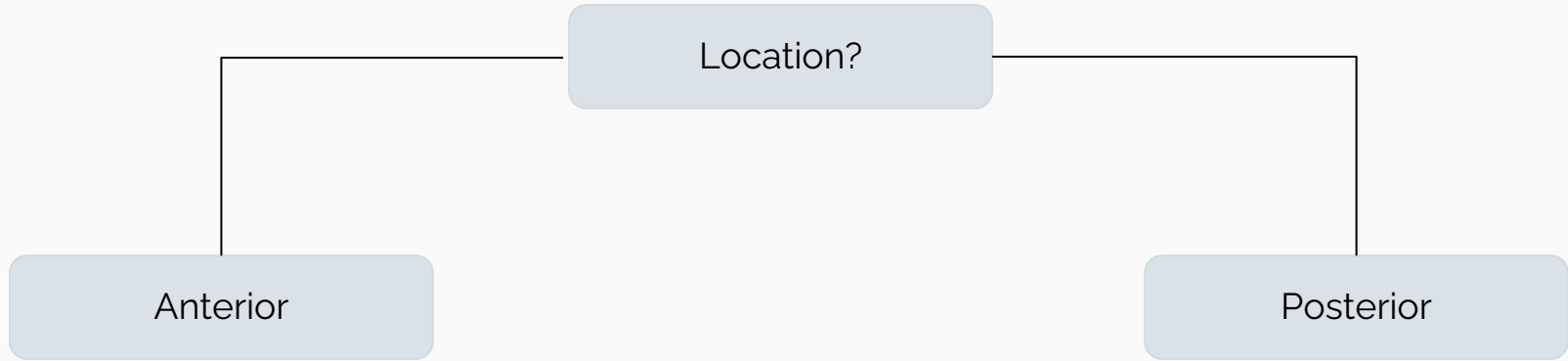
Directly after curing composite



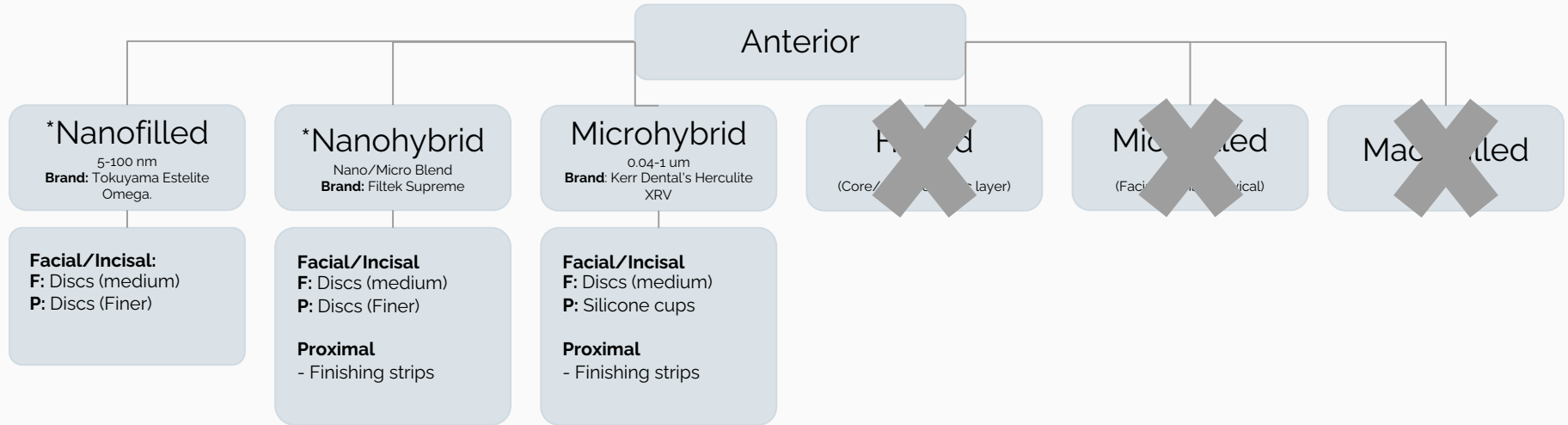
After finishing & polishing



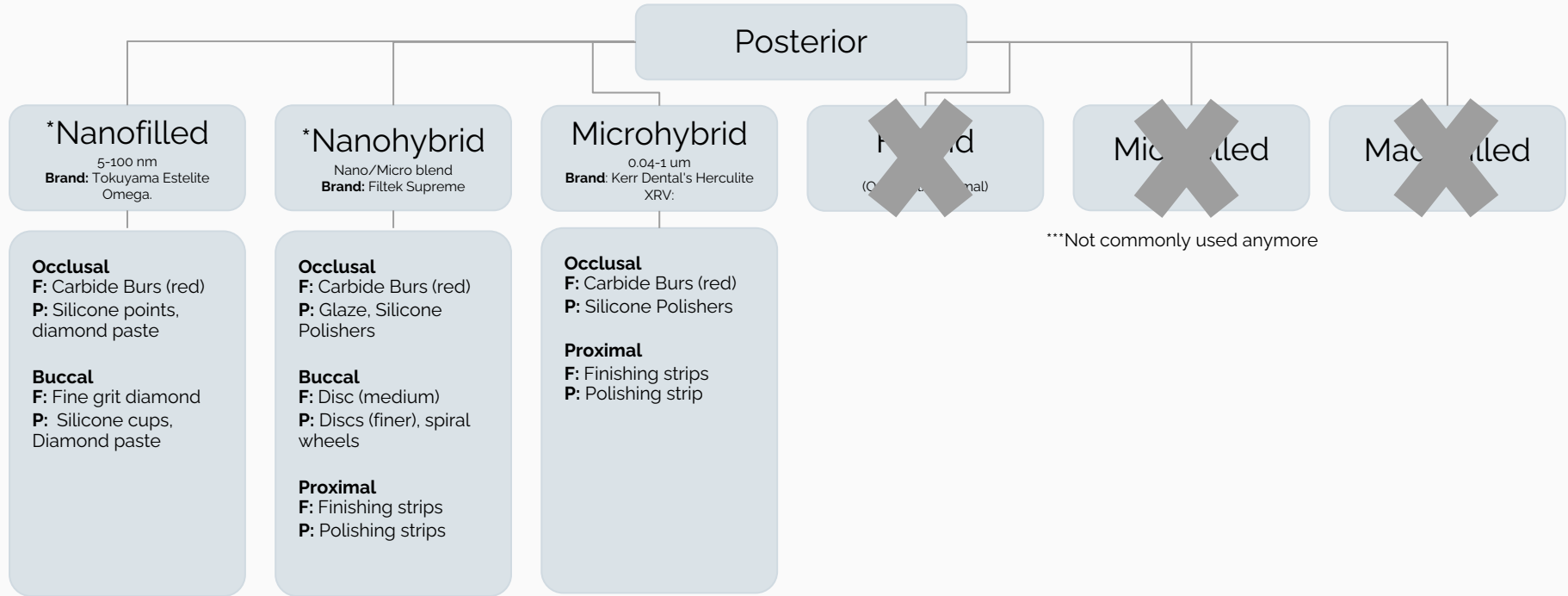
Decision Tree



Decision Tree



Decision Tree



Behavioral Objectives Review

BO1: Explain surface roughness, how it is measured, and its effect on the quality and glossiness of a composite restoration.

Surface roughness is the **irregularities, deviations, or unevenness** on the material surface from its mean value due to cutting forces. It is measured by **Ra, Rz, and Rq values** or Sa/Sq values. **Increased surface roughness correlates to increase dullness, surface staining/color/discoloration, plaque accumulation (Threshold Ra = 0.2um), and abrasion/adhesion.** It is also associated with decreased microhardness, decreased gloss and patient discomfort.

Behavioral Objectives Review

BO2: Describe the different steps for achieving ideal surface contour & smoothness. Identify the different types of smoothing systems used for posterior composites.

The sequence of achieving composite smoothness includes finishing (shape correction to match tooth anatomy) then polishing (eliminating scratches to create a smoother surface). Carbide burs and coarse discs are used for finishing. Polishing tools include one-step enhance points or multi-step polishing discs, or polishing pastes. While glazes for dental composites are on the market, high-level evidence for it is limited due to reports of early wear or debonding.

Behavioral Objectives Review

BO 3: Describe the relationship between filler content and overall composite surface wear/longevity

Among nanofilled, nanohybrid, microhybrid, and microfilled resins, variations in filler type, size, shape, and loading determine both surface roughness and long term durability. As composites advanced from macrofilled to nanofilled systems, smaller and more spherical fillers with stronger resin coupling created smoother surfaces, less wear, and better gloss retention.

Surface wear and longevity are influenced by all filler characteristics. Smaller and more uniform fillers resist plucking and maintain smoothness, spherical shapes promote even stress distribution and easier polishing, and higher filler loading increases hardness and limits resin exposure to abrasion. Together, these factors explain why modern nanofilled and nanohybrid composites provide the best balance of esthetics, wear resistance, and long term clinical performance.

Behavioral Objectives Review

BO #4: Describe the role of finishing and polishing systems in decreasing surface roughness of different composite types from immediate placement to ensuring long term smoothness and longevity.

The smoothest surface finishes were achieved when Esthet-X HD (microhybrid) was polished with Sof-Lex fine and superfine discs, Ceram X Mono (nanohybrid) with tungsten carbide multi-blade cutters followed by Sof-Lex, and Filtek Silorane (silorane-based microhybrid) with Sof-Lex discs and silicone rubber tips. The roughest surfaces were observed in samples that were unpolished or finished with diamond burs only.

Behavioral Objectives Review

BO #4: Describe the role of finishing and polishing systems in decreasing surface roughness of different composite types from immediate placement to ensuring long term smoothness and longevity (CONTINUED)

While glossiness and surface roughness of a direct restoration is determined by polishing systems during placement, **long term longevity of a restoration is determined by the filler content due to the size of the particles. Irregular fillers wear faster** while spherical and rhomboid silica and zirconia fillers tend to show more resistance. The **level of hydrophobicity also impacts wear resistance.**

Behavioral Objectives Review

BO5: Explain the maintenance strategies and repair techniques for composite restorations.

Maintenance of the restoration:

Maintaining the **esthetics, function, and integrity** of composite restorations is essential to reduce the need for repair or replacement, which most often result from **fracture or secondary caries**. This responsibility is shared among the **dentist, dental hygienist, and patient**. The dentist routinely evaluates restorations and keeps staff informed about materials and techniques. The dental hygienist uses **proper polishing methods** to promote long-term success, while the patient practices **good oral hygiene**, follows recommended products and techniques, and understands their caries risk and ways to reduce it.

Behavioral Objectives Review

BO5: Explain the maintenance strategies and repair techniques for composite restorations (CONTINUED)

Repair:

When deciding whether to repair or replace a composite restoration, clinicians should consider time, resources, remaining tooth structure, restoration longevity, and pulpal health. **Repair is preferred when it preserves tooth structure, extends the restoration's life, and minimizes pulpal trauma, especially for posterior teeth that can be monitored until plaque accumulation occurs** (Blum et al., 2011).

Air abrasion with **aluminum oxide** provides the **highest repair strength** by creating superior micro-retentive interlocking between old and new composite surfaces (da Costa et al., 2012).

References

BO1



BO2



BO3



BO4



BO5



References

Case Studies

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Contouring and Polishing Anterior Resin Composite Restorations:
A Checklist Manifesto. Clinician's Choice Dental Products Inc.

<https://www.clinicianschoice.com/systemic-approach-contouring-polishing-anterior-resin-composite-restorations-checklist/>

6 Thinking Hats (Kahoot)



White Hat (Faculty): Provides objective data and scientific evidence.



Red Hat (Patient): Expresses emotions, concerns, and preferences.



Black Hat (Dental Insurance): Evaluates risks, coverage limitations, and cost concerns.



Yellow Hat (Dentist): Highlights the benefits, clinical advantages, and positive outcomes.



Green Hat (Manufacturer): Explores innovation, future possibilities, and improvements.



Blue Hat (Board of Dentistry): Ensures all perspectives are considered, standards and ethical guidelines.

Thank you!